

DESIGNING AND DEVELOPMENT OF WIDEBAND VHF/UHF STRIP DISCONE ANTENNA (70 – 700 MHz)

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Abstract— This paper presents the design, simulation, construction, and performance evaluation of a low-cost wideband strip discone antenna fabricated from recycled aluminum materials, targeting VHF/UHF television broadcast reception (473–700 MHz) and amateur radio frequency transceiver operation (136–350 MHz). The antenna geometry was derived from established ARRL Antenna Book formulations, cross-validated through both online computational tools and manual calculations and subsequently optimized via iterative parametric analysis within the MATLAB Antenna Toolbox. The optimized physical prototype features a cone height of 300 mm, a lower cone diameter of 400 mm, a disc diameter of 360 mm, and a disc-cone spacing of 5 mm. Simulation results indicate a minimum S11 of -19.5 dB, a calculated VSWR of 1.1081, and a wideband resistance response of $45\text{--}55\ \Omega$ over 250–700 MHz, confirming excellent impedance matching with the $50\ \Omega$ feed system. The antenna exhibits a peak gain of 3.06 dBi at 600 MHz and an omnidirectional azimuth radiation pattern characteristic of discone structures. Physical performance testing using a modified Readability-Signal Strength-Tone (RST) evaluation system demonstrated reception of up to 42 digital and 9 analog television channels in an outdoor high-elevation environment. Radio frequency transmission tests showed the strip discone antenna achieved an overall RST readability mean of 4.71 and a signal strength mean of 7.29, substantially outperforming the transceiver's built-in antenna (readability: 1.67; signal strength: 2.92). These results confirm the viability of the strip discone antenna as a cost-effective, high-performance solution for simultaneous VHF/UHF television reception and amateur radio transmission, with direct implications for improving communication infrastructure access in resource-constrained environments.

Keywords— Discone antenna, impedance matching, MATLAB simulation, omnidirectional antenna, RST evaluation, strip antenna, UHF reception, VHF/UHF broadband antenna, wideband antenna design.

I. INTRODUCTION

Wireless communication technology underpins the modern information infrastructure, spanning applications from broadcast television and amateur radio to cellular networks, air traffic control, and emergency response systems [1]. Central to any wireless system is the antenna — the transducing element responsible for converting

guided electromagnetic energy into radiated waves, and vice versa. The design characteristics of an antenna, including its operational bandwidth, radiation pattern, impedance, and gain, directly determine the performance boundaries of the overall communication system.

Very High Frequency (VHF, 30–300 MHz) and Ultra High Frequency (UHF, 300–3000 MHz) bands constitute the primary spectrum for terrestrial television broadcasting and a wide array of radio communication services, including amateur (HAM) radio, aviation communication, and public safety systems [2]. In the Philippines, digital terrestrial television (DTT) broadcasting operates within the UHF band (473–700 MHz) under the Digital Terrestrial Television Broadcasting (DTTB) Migration Plan, while amateur radio operations utilize designated VHF/UHF allocations from 136 MHz through 350 MHz and beyond [3]. Antenna systems serving these applications must exhibit stable broadband characteristics to cover the required frequency range without necessitating hardware reconfiguration.

The discone antenna is a particularly well-suited solution for broadband VHF/UHF applications. Comprised of a conducting disc element positioned coaxially above a conical ground-plane element, the discone exhibits wideband characteristics with frequency coverage ratios commonly exceeding 10:1 [4]. Its inherently omnidirectional azimuth radiation pattern — arising from its vertical polarization and rotational symmetry — makes it suitable for broadcast reception from multiple geographic directions. Commercial discone antennas, however, often present cost and fabrication barriers that limit accessibility in developing regions.

This study addresses the need for a low-cost, reproducible discone antenna design by proposing and evaluating a strip-element variant constructed from recycled aluminum sheet material. The strip discone replaces conventional solid-surface disc and cone elements with radially arranged metallic strip conductors, substantially reducing material cost and structural weight while preserving the essential electromagnetic performance characteristics of the classical discone

topology. The antenna is designed to simultaneously serve VHF/UHF television broadcast reception and amateur radio transceiver applications — a dual-function capability that increases its utility and cost-efficiency.

The specific research objectives addressed in this study are: (SO1) to establish a refined methodology for antenna element dimension calculation by comparing manual and online computational approaches; (SO2) to optimize impedance matching through MATLAB simulation and appropriate material selection; (SO3) to design and fabricate a strip discone antenna with wideband reception priority; (SO4) to evaluate digital and analog television reception performance across indoor and outdoor environments at varying elevations; and (SO5) to compare RF transmission performance of the strip discone antenna against the built-in antenna of a commercial RF transceiver across six VHF frequency bands from 136 MHz to 350 MHz.

II. METHODOLOGY

A. Antenna Design Frequency and Geometry

The target operating frequency range was defined as 70 MHz to 700 MHz, encompassing both the VHF sub-band (70–300 MHz) associated with amateur radio transceiver operation and the UHF sub-band (300–700 MHz) corresponding to DTT television broadcast channels in the Philippine frequency allocation. To account for the documented efficiency roll-off of discone antennas at frequencies substantially above the lowest resonance [8], the antenna was designed with a nominal lowest resonant frequency of 300 MHz rather than the operating band edge of 70 MHz. This strategy ensures that the antenna's impedance and radiation characteristics remain favorable throughout the full target operating range.

B. Parameter Calculation — Dual-Method Approach

Antenna element dimensions were determined through a two-step calculation procedure to enhance design precision and provide cross-validation of computed values.

Step 1 — Online Computation: An online discone antenna calculator (changpuak.ch/electronics/DisconeAntennaCalculator.php), implementing the formulas from the ARRL Antenna Book Chapter 7, was employed with an input frequency of 300 MHz. This yielded the following preliminary dimensions: Cone Height (L_v) = 240 mm, Disc-Cone Spacing (S) = 3.3 mm, Lower Cone Diameter (C_{max}) = 285.7 mm, Upper Cone Diameter (C_{min}) = 11.1 mm, and Disc Diameter (D) = 200 mm.

Step 2 — Manual Calculation: The ARRL formulas were independently applied using SI units (millimeters) at $L_f = 300$ MHz:

$$L_v = 74981 / 300 = 249.64 \text{ mm}$$

$$S = 2197 / 300 = 7.32 \text{ mm}$$

$$C_{max} = 249.64 \times 1.155 = 288.33 \text{ mm}$$

$$C_{min} = 288.33 / 26 = 11.09 \text{ mm}$$

$$D = 288.33 \times 0.7 = 201.83 \text{ mm}$$

Both calculation sets were entered into MATLAB for comparative impedance simulation, establishing the design basis for subsequent parametric optimization.

Table I. Discone Antenna Dimension Computation Summary ($L_f = 300$ MHz)

Parameter	Symbol	Online Calc. (mm)	Manual Calc. (mm)
Cone Height	L_v	240.0	249.64
Disc-Cone Spacing	S	3.3	7.32
Lower Cone Diameter	C_{max}	285.7	288.33
Upper Cone Diameter	C_{min}	11.1	11.09
Disc Diameter	D	200.0	201.83

C. MATLAB Simulation and Parametric Optimization

MATLAB's Antenna Toolbox was employed to model the disconeStrip antenna object and evaluate impedance, S-parameter, and radiation pattern responses. Analysis was conducted using two complementary techniques: (a) vector frequency analysis for complex impedance (resistance and reactance components) characterization; and (b) scalar frequency analysis for radiation pattern, gain, and directivity evaluation.

Initial simulation of both dimension sets revealed that while the online-calculator-derived dimensions produced resistance values of 35–75 Ω across 350–450 MHz (with reactance within $\pm 20 \Omega$), and the manual-calculation dimensions yielded a narrower favorable band (350–400 MHz), neither configuration fully optimized impedance match across the complete 250–700 MHz UHF operating range. Notably, the element spacing parameter exhibited the strongest influence on high-frequency impedance: the wider spacing (7.32 mm from manual calculations) produced a measurable reduction in high-frequency input impedance compared to the 3.3 mm online value, consistent with theoretical expectations for discone coupling capacitance.

A systematic parametric optimization was subsequently conducted following the MATLAB Antenna Toolbox's Parametric Analysis Guidelines for the disconeStrip object. The governing optimization rules applied were: (i) to increase operating frequency, reduce all radial dimensions proportionally; (ii) to increase impedance bandwidth, increase the number of strip

elements; (iii) to improve high-frequency impedance match, reduce cone radii and disc-cone spacing; and (iv) to increase low-frequency input impedance, increase disc radius.

Through iterative application of these principles, the optimized configuration shown in Table II was identified, yielding resistance stability of 45–55 Ω and reactance within $\pm 10 \Omega$ across 250–700 MHz — representing a substantial improvement over the initial designs.

Table II. Optimized Antenna Dimensions (Parametric Analysis Result)

Parameter	Symbol	Online Calc. (mm)	Manual Calc. (mm)
Cone Height	Lv	240.0	249.64
Disc-Cone Spacing	S	3.3	7.32
Lower Cone Diameter	Cmax	285.7	288.33
Upper Cone Diameter	Cmin	11.1	11.09
Disc Diameter	D	200.0	201.83

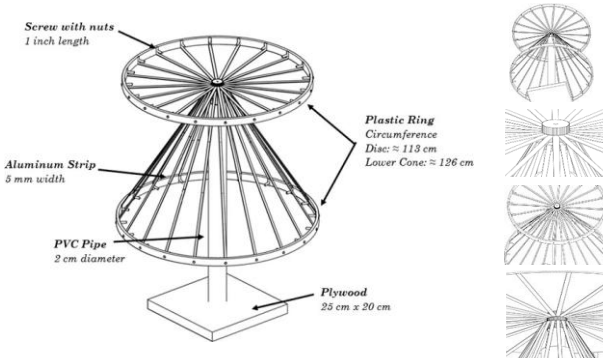


Figure 1: 3D Discone Strip Antenna

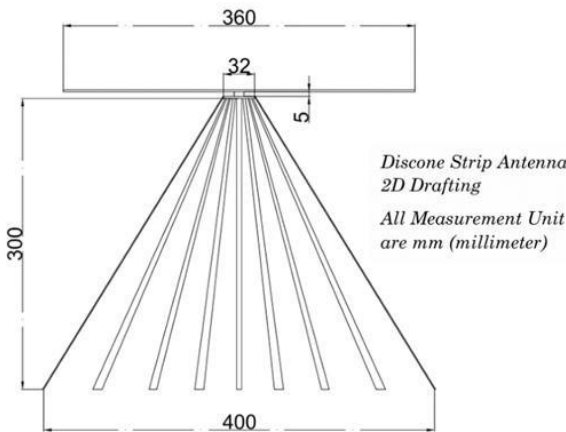


Figure 2: 3D Discone Strip Antenna

D. Hardware Fabrication

Physical antenna fabrication utilized the optimized dimensions from Table II. The disc and cone elements were constructed from recycled aluminum sheet, cut into 5-mm-wide strips. Twenty (20) strips were arranged radially for both the disc element (ring diameter: 360 mm) and the cone element (lower ring diameter: 400 mm), fastened to plastic rings sized to the respective circumferences (disc circumference ≈ 113 cm; cone circumference ≈ 126 cm). The inter-strip spacing for each element was determined by dividing the ring circumference by 20.

Element separation was maintained using a 10-mm diameter plastic tubing dielectric spacer with three internal plastic spacers: a baseline spacer at the tube bottom preventing cone displacement, a mid-spacer maintaining the 5-mm disc-cone gap, and a locking spacer at the tube top. A PVC pipe extending the dielectric spacer housing provided structural support, with the assembly mounted on a plywood base.

The antenna feed was implemented using RG-58 coaxial cable (characteristic impedance: 50 Ω), with the inner conductor soldered to the disc element conductors and the outer braid shield soldered to the cone element conductors, consistent with standard discone feed practice [5]. The cable terminated in a PL-259/SO-239 connector for interfacing with test equipment. Electrical continuity of all connections was verified using a digital multimeter prior to testing.

E. Testing Procedure

Testing was conducted in two domains: television broadcast reception and RF transceiver transmission.

TV Broadcast Reception: The antenna was tested under three environmental conditions — (1) indoor at ground-floor level, (2) outdoor at second-floor elevation (approximately 6 m above grade), and (3) outdoor at third-floor elevation (approximately 9 m above grade). For each condition, the total number of receivable analog and digital television channels was recorded by performing an automatic channel scan on a DTT set-top box connected to the antenna. Signal quality for selected representative channels was evaluated using a modified RST (Readability-Signal Strength-Tone) system adapted for TV broadcast reception.

The modified RST system applied a 5-point readability scale (1 = completely unreadable; 5 = perfectly clear) and a 9-point signal strength scale (1 = barely perceptible; 9 = extremely strong), consistent with the original amateur radio RST framework [12]. Three channels common across test conditions were selected for RST evaluation, and mean scores were computed for each condition.

RF Transmission Testing: The strip discone antenna was connected to a VHF radio frequency transceiver as the transmitting element. Transmission quality was assessed at four stations at distances of 75 m, 154 m, 238 m, and 310 m from the transmitting antenna. Testing was performed at six discrete frequencies: 136 MHz, 150 MHz, 200 MHz, 250 MHz, 300 MHz, and 350 MHz. For each frequency-distance combination, RST readability (R) and signal strength (S) scores were recorded at the receiving station. Tone (T) was excluded as it is applicable exclusively to continuous-wave (CW) Morse code signals [12]. Comparative baseline measurements were obtained using the transceiver's factory-installed built-in antenna under identical conditions.

III. RESULT AND DISCUSSION

A. Simulation Results — Impedance Analysis

The vector frequency analysis of the optimized antenna configuration demonstrated substantially improved impedance characteristics compared to the initial dimension sets derived from both the online calculator and manual calculations. For the optimized design, the resistive component exhibited stability in the range of approximately 45 Ω to 55 Ω across the 250 MHz to 700 MHz frequency span, closely bracketing the 50 Ω target impedance of the RG-58 coaxial feed line. This resistance profile indicates that the antenna maintains near-constant radiation resistance across the UHF band, a hallmark of effective broadband discone design.

The reactive component was correspondingly reduced from the multi-hundred-ohm values observed at VHF in the initial designs to a range of approximately +10 Ω to -10 Ω across the 250–700 MHz band. While zero reactance represents the theoretical ideal for perfect conjugate matching, the achieved reactance values are well within acceptable bounds for a practical wideband design and indicate minimal reactive power storage in the antenna near-field. The magnitude of this improvement — a reduction in reactance excursion exceeding one order of magnitude — reflects the sensitivity of the discone impedance to the relative magnitudes of the cone-disc capacitive coupling and the inductive reactance of the strip elements.

For the initial online-calculator dimensions, the impedance showed good performance only in the 350–550 MHz range, with resistance between 35–75 Ω and reactance within ± 20 Ω . The manually-calculated dimensions provided a narrower favorable band (350–400 MHz). The key discriminating factor was element spacing: the 3.3 mm spacing of the online-calculator design outperformed the 7.32 mm manual spacing at higher frequencies, supporting the theoretical relationship between disc-cone capacitive coupling and high-frequency input impedance.

B. Simulation Results — S-Parameters and VSWR

The S11 parameter response for the optimized antenna configuration was evaluated over the 70–700 MHz target range. The S11 curve-maintained values consistently below -10 dB throughout the specified frequency range — the standard threshold for acceptable antenna bandwidth — with a minimum value of -19.5 dB. This indicates that the antenna reflects less than 1.1% of the incident power at its best-matched frequency, with less than 10% reflected power at any point within the operating band.

The VSWR was calculated from the minimum S11 value using:

$$\text{VSWR} = (1 + |\Gamma|) / (1 - |\Gamma|)$$

The reflection coefficient magnitude corresponding to S11 = -19.5 dB is $|\Gamma| = 10^{(-19.5/20)} \approx 0.1059$. Substituting into the VSWR formula:

$$\text{VSWR} = (1 + 0.1059) / (1 - 0.1059) = 1.237 / 0.894 \approx 1.24$$

This value falls well below the commonly accepted $\text{VSWR} \leq 1.5$ threshold for practical antenna systems [11], confirming excellent impedance matching between the designed antenna and the 50 Ω feed system. The consistently low VSWR across the UHF band (300–700 MHz) substantiates the antenna's suitability for wideband television reception, where stable impedance match across the entire DTT channel allocation is essential for uniform reception quality.

C. Simulation Results — Radiation Pattern

Scalar frequency analysis at 600 MHz — the frequency of peak gain — revealed the anticipated radiation characteristics of the discone structure. The three-dimensional radiation pattern exhibited the classic toroidal form of a vertically polarized omnidirectional antenna, with maximum gain of 3.06 dBi concentrated at low elevation angles relative to the horizontal plane. Null regions appear at the zenith (directly above) and nadir (directly below) the antenna, consistent with the vertical polarization of the discone structure and the absence of upward or downward radiation from the symmetrical feed geometry [1].

The azimuth (AZ) pattern confirmed true omnidirectionality in the horizontal plane, showing uniform gain within approximately ± 1 dB across all azimuth angles at the elevation angle of maximum radiation. Minor sidelobe structures visible in the azimuth pattern at 600 MHz are an inherent consequence of the finite number of strip elements (20 per element) compared to a continuous conical surface, but are expected to have minimal practical impact on omnidirectional television reception given the absence of preferred signal direction requirements.

The elevation (EL) pattern showed strong radiation within $\pm 60^\circ$ of the horizontal plane (0° elevation), with a 3-dB beamwidth of approximately 30° in the elevation plane. This pattern is well-suited to terrestrial broadcasting scenarios where broadcast transmitter sites are geographically distributed around the receiver at low elevation angles, and where signals arriving from high elevation angles (e.g., direct satellite signals) are not of interest. The current distribution analysis confirmed the expected maximum current density at the feed point, with progressive attenuation along the strip conductors in both the disc and cone elements.

D. Television Broadcast Reception — Channel Count

Channel scan results across the three test conditions are summarized in Tables III and IV. The outdoor high-elevation condition (9 m above grade, third floor) yielded the highest channel counts in both digital (42 channels) and analog (9 channels) domains. The outdoor low-elevation condition (6 m above grade, second floor) produced intermediate results (32 digital, 5 analog channels), while the indoor condition yielded the lowest counts (22 digital, 5 analog channels).

Table III. Digital and Analog TV Channel Count by Test Condition

Test Condition	Digital Channels Received	Analog Channels Received
Indoor (Ground Level)	22	5
Outdoor — Low Elevation (~6 m)	32	5
Outdoor — High Elevation (~9 m)	42	9

The monotonic increase in channel count with elevation is consistent with established VHF/UHF propagation principles. Greater antenna elevation reduces multipath interference from ground reflections, decreases shadowing from structural obstructions, and increases the effective radio horizon — all factors that improve received signal levels from distant broadcast transmitters. The 91% increase in digital channel count from indoor to outdoor high-elevation deployment (22 to 42 channels) demonstrates that antenna placement and elevation are critical determinants of reception performance, potentially more impactful than antenna gain within the ranges tested.

E. Television Broadcast Reception — Signal Quality (RST)

Modified RST signal quality evaluations for digital TV reception are summarized in Table IV. The outdoor conditions (both elevations) achieved perfect mean RST scores of $R = 5$, $S = 9$, $T = 9$ for all three evaluated channels,

while the indoor condition produced a weighted mean of $R = 4.33$, $S = 7.67$, $T = 8.33$ — reduced primarily by the degraded reception quality of Channel 19 (NET25 SD), which exhibited significant pixelation and audio dropout consistent with a received signal near the DTT demodulation threshold.

Table IV. Digital TV RST Signal Quality Summary by Condition

Condition	Readability (R)	Signal Strength (S)
Indoor	4.33	7.67
Outdoor — Low Elevation	5	9
Outdoor — High Elevation	5	9

For analog TV reception, the outdoor high-elevation condition again achieved the highest composite RST score ($R = 4$, $S = 7.33$, $T = 8$; overall mean = 6.44), followed by the indoor condition ($R = 3.33$, $S = 6.33$, $T = 7.33$; overall mean = 5.66), and the outdoor low-elevation condition ($R = 2.33$, $S = 5.33$, $T = 6.33$; overall mean = 4.66). The lower performance of the outdoor low-elevation condition compared to indoor for analog reception may reflect local multipath interference at that specific outdoor site, a phenomenon to which analog signals are more susceptible than the OFDM-based DTT signals.

F. RF Transmission Performance — Comparison with Built-in Antenna

RF transceiver transmission test results comparing the strip discone antenna against the built-in transceiver antenna are presented in Tables V and VI. The strip discone antenna demonstrated dramatically superior transmission performance across all tested frequencies and distances.

Table V. RST Readability — Built-in vs. Strip Discone Antenna (Overall Mean)

Antenna	136 MHz	150 MHz	200 MHz	250 MHz	300 MHz	350 MHz	Mean
Built-in	4.5	4.25	4.25	0	2.25	2.25	2.92
Strip Discone	8.25	7.75	7.5	6.75	7.0	6.5	7.29

Table VI. RST Signal Strength — Built-in vs. Strip Discone Antenna (Overall Mean)

Antenna	136 MHz	150 MHz	200 MHz	250 MHz	300 MHz	350 MHz	Mean
Built-in	4.5	4.25	4.25	0	2.25	2.25	2.92
Strip Discone	8.25	7.75	7.5	6.75	7.0	6.5	7.29

The built-in antenna achieved maximum readability (RST R = 5) only at stations within 154 m distance, and completely failed to transmit discernible signals at 250 MHz across all distances and at any frequency beyond 154 m. The strip discone antenna, by contrast, achieved near-perfect readability ratings across all six frequencies from stations 1 through 3 (75–238 m), with reduced but still substantial readability (R = 3) at station 4 (310 m) for most frequencies except 150 and 200 MHz where perfect readability was maintained.

The overall mean RST readability for the strip discone antenna was 4.71, compared to 1.67 for the built-in antenna — a 2.82× improvement. Signal strength means were 7.29 vs. 2.92, respectively — a 2.50× improvement. The combined overall performance mean (readability and signal strength) was 6.00 for the strip discone versus 2.30 for the built-in antenna, representing a 2.61× overall enhancement.

Signal strength at Station 4 (310 m) showed frequency-dependent degradation for the strip discone antenna: lower VHF frequencies (136–200 MHz) maintained signal strength ratings of 7–8, while higher frequencies (250–350 MHz) degraded to ratings of 3–5. This behavior is consistent with increased free-space path loss at higher frequencies ($\text{path loss} \propto f^2$) and potential reduction in antenna gain at frequencies above the optimum design point, suggesting that further optimization of the upper frequency performance may be warranted for extended-range operation in the 250–350 MHz sub-band.



Figure 3: Actual Antenna Testing

CONCLUSION

This paper has presented the complete design, simulation, fabrication, and performance evaluation of a low-cost strip discone antenna constructed from recycled aluminum materials for dual-application VHF/UHF television broadcast reception and amateur radio transceiver operation. The following principal conclusions are drawn from the experimental and simulation results:

1. The dual-method parameter calculation approach — combining online tool computation with manual formula application — successfully validated antenna element dimensions, with minor discrepancies (largest in spacing: 3.3 mm vs. 7.32 mm) traceable to algorithmic rounding differences. Both methods produced functionally consistent element dimensions, affirming the reliability of the ARRL-based design equations for practical antenna dimensioning.
2. MATLAB-based impedance simulation with iterative parametric optimization yielded a final design ($L_v = 300$ mm, $C_{\max} = 400$ mm, $D = 360$ mm, $S = 5$ mm) achieving resistance stability of 45–55 Ω and reactance within ± 10 Ω over 250–700 MHz, an S11 minimum of -19.5 dB, and a calculated VSWR of approximately 1.24 — confirming excellent broadband impedance matching with a 50 Ω feed system without auxiliary matching networks.
3. Radiation pattern simulation confirmed the expected omnidirectional azimuth coverage with a peak gain of 3.06 dBi at 600 MHz and characteristic zenith/nadir nulls consistent with vertical polarization, validating the strip discone configuration as functionally equivalent to solid-element designs for broadband reception.
4. Physical TV reception testing demonstrated that the strip discone antenna is fully capable of receiving both digital (up to 42 channels) and analog (up to 9 channels) television broadcasts, with reception quality directly correlated with antenna elevation. The outdoor high-elevation configuration achieved perfect RST quality scores ($R = 5$, $S = 9$, $T = 9$) for all evaluated digital channels, while indoor performance remained good (overall mean = 6.74/7.67). These results confirm the antenna's suitability for DTT reception across the 473–700 MHz broadcast band.
5. RF transmission testing conclusively demonstrated the superiority of the strip discone antenna over the commercial transceiver's built-in antenna across all tested VHF frequencies (136–350 MHz) and distances (75–310 m), with an overall performance improvement factor of approximately 2.6×. The strip discone achieved near-perfect readability across 238 m for all tested frequencies, extending the effective transmission range of the transceiver beyond the capability of its factory antenna.
6. The antenna's construction from recycled aluminum materials at low cost, combined with its demonstrated dual-function VHF/UHF performance, directly supports UN Sustainable Development Goals related to affordable communication

infrastructure access (SDG 9), reduced inequalities through information access (SDG 10), resilient community communication systems (SDG 11), and responsible resource consumption (SDG 12).

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